Yogita

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	9.04
NDMC NAVYUG SCHOOL LAXMI BAI NAGAR NEW DELHI	Class XII (CBSE)	2017	93.60%
NDMC NAVYUG SCHOOL LAXMI BAI NAGAR NEW DELHI	Class X (CBSE)	2015	10.00%

WORK EXPERIENCE

Junior Architect - *InnovArch, Noida*

- **Led sustainable retrofit planning** for **Physics Wallah educational facilities** , aligning upgrades with **NBC 2016**
- Focused retrofit upgrades on improving functionality, lifecycle performance, and building aesthetics.
- **Designed sustainable site infrastructure**, integrating **rainwater harvesting and wastewater drainage systems.**
- **Developed architectural and construction drawings** from **72,000 sq. ft. master planning** to **400 sq. ft. residential units.**

Junior Architect - *AE Studio, New Delhi*

- **Contributed to the architectural design team** for **Chintamanis High-Rise Residential Development (8.63 acres)**, a project that achieved **IGBC Platinum Pre-Certification,** while ensuring compliance with **NBC standards and Gurugram municipal bylaws.**
- **Coordinated MEP, structural, landscape, and civil teams** for multidisciplinary integration in sanctioned and execution drawings.
- **Evaluated low-carbon construction techniques** across residential and commercial projects
- Developed **construction details, bill of quantities (BOQ), 3D visualizations, and client presentations** for these projects.
- **Led and mentored 3-4 architectural interns**, strengthening team coordination.
- Supported **corporate social media and portfolio communications,** increasing project visibility.

Architectural Intern - *Babbar & Babbar Architects (BBArch), New Delhi*

- **Developed eco-resort master planning and site design** for Ramakrishna Mission, across **5-acre hilly terrain**, using **contour-responsive** planning and **low-impact cut-and-fill strategies.**
- **Evaluated natural materials and prefabricated construction systems** for **sustainable, climate-resilient development.**
- **Prepared climate-responsive** landscape construction drawings and material specifications for **GHWP Water Park, Dubai,** considering **solar exposure and arid climate condition and site-specific material requirements.**

Intern - *CEEW (Council on Energy, Environment & Water), New Delhi*

- **Conducted spatial surveys and urban mobility analysis** for a Residential Parking & Traffic Management Project, mapping roads, parking, utilities, tree cover, and circulation using **AutoCAD** to improve space utilization and emergency access.
- **Assessed EV charging feasibility and proposed sustainable street redesigns**, integrating **EV infrastructure, traffic calming, pedestrian/cycling safety, and active mobility** to support data-driven urban sustainability planning.

PROJECTS

Fluoride Removal Using Waste-Derived Calcium-Based Adsorbent | *M.Tech Thesis*

Working under the guidance of Dr. Ambika S

- **Developed a waste-to-resource fluoride-removal material** by processing a mineral-rich waste stream, including cleaning, thermal/chemical treatment, particle-size control, and adsorption testing for contaminated water.
- Synthesized and characterized a **calcium carbonate (CaCO)-based adsorbent** through a precipitation route; **FTIR**

confirmed CaCO formation and supported the calcite phase, establishing an optimization baseline.
- Evaluated fluoride removal via **Ion Chromatography (IC)**, establishing baseline performance of the untreated material.
- Currently optimizing activation and adsorption performance, followed by sustainability analysis of the treatment system.

Comparative TEA of Battery Energy Storage Systems (BESS) |*Sustainable Energy Technology*

- **Collaborated on NREL SAM simulations** comparing Lithium Iron Phosphate (LFP) and Nickel Cobalt Manganese (NCM) battery systems.
- **Evaluated Levelized Cost of Energy (LCOE), Net Present Value (NPV), and Internal Rate of Return (IRR)**.
- Assessed energy throughput and degradation for lifecycle performance and financial feasibility.

Route Optimization for Sustainable Mobility in Pune |*Analysis and Methods for Sustainable Engineering*

- **Developed a multi-objective optimization model** in **LINGO** to allocate diesel and electric buses across **8 routes**, minimizing operating cost, energy consumption, and carbon emissions while maximizing passenger service.
- Analyzed route-level demand, capacity, travel time, and fleet constraints to support data-driven sustainable transport planning.

Sensitivity Analysis of Concrete Mixes using Life Cycle Assessment (LCA) | *Sustainable Building Materials*

- **Conducted LCA** using the **ReCiPe Midpoint (H)** method in **openLCA** to quantify environmental impacts across multiple categories.
- Performed **+1% sensitivity analysis** to identify **Ordinary Portland Cement (OPC)** as the dominant impact driver.
- Assessed the role of **Supplementary Cementitious Materials (SCMs)** in concrete mix environmental impacts.

Scope 1 GHG Assessment - *IIT Hyderabad*

- Conducted **Scope 1 GHG** assessment covering boundary definition, activity-data collection, emission-factor selection, and emissions quantification
- Quantified **1,136.59 tCOe/year** from stationary combustion, identifying hotspots across diesel generators, LPG cooking, and waste incineration
- Evaluated mitigation scenarios rooftop solar, kitchen electrification/biogas, waste management, energy efficiency estimating a potential **540 tCOe/year (50%)** reduction

Global Warming Potential -Experimental Study | *Carbon Capture, Utilization and Storage*

- Designed a 3-setup experiment comparing infrared heat absorption and retention of **CO, CH, and N** under varying radiation.
- **Validated GWP characterization factors** experimentally using an LDPE pouch and IR heater setup (2-10 µm); found CO retained heat longest and prepared a structured report covering methodology, observations, limitations, and conclusions.

Sustainable Village Tourism - *Hampi*

- Assessed Hampi, Karnataka, as a sustainable-tourism case study, addressing tourism-driven energy demand, waste generation, sanitation, and infrastructure pressures through renewable energy, waste-to-energy, recycling, and **circular-economy** strategies.
- Evaluated the **LCA** of a **250 kWp bifacial TOPCon solar PV system** and **Deenbandhu biogas plant** using **openLCA**, analyzing carbon emissions, energy use, resource consumption, and environmental hotspots across defined lifecycle stages.

Adaptive Reuse of Urban Infrastructure for Decentralized Farming | *Concepts in Sustainable Engineering*

- **Evaluated 12 Indian and international case studies on decentralized farming** using environmental, spatial, technological, and socio-economic indicators, including energy consumption, water use, **GHG emissions**, and yield.
- Proposed **adaptive-reuse strategies** for residential, institutional, and commercial infrastructure integrating water conservation, renewable energy, waste circularity, and GHG reduction, and mapped project impacts to relevant **UN SDGs**.

Sustainable Cotton Farming: Strategies to Minimize Water Consumption |*Concepts in Sustainable Engineering*

- Assessed the **water footprint** of cotton cultivation in Telangana and quantified water requirements across crop growth stages.
- Evaluated **drip irrigation**, **alternate furrow irrigation**, **sensor-based deficit irrigation**, treated wastewater reuse, and drought-tolerant varieties, estimating a **~60% reduction** in water footprint through combined strategies.

ACHIEVEMENTS

3rd Rank, nVisionX'26 - IIT Hyderabad × GHMC Design Challenge | *Team RouteMind*

- Solved **heterogeneous capcitated vehicle-routing** problem for Greater Hyderabad
- Analyzed environmental and operational data for **1,583 garbage-vulnerable points** and **18 transfer stations** across 650 sq. km, Hyderabad
- Developed a data-driven framework to reduce travel distance, fuel consumption, and associated **emissions**

IYBC 2026 | *Telangana Biodiversity Board*

- Selected as **one of 100 delegates** for the International Youth Biodiversity Conference (IYBC) 2026 at Shanti Vanam.
- Secured **1st position** in cultural team competitions.

Smart India Hackathon (SIH) 2026 - Internal Hackathon Round Qualifier | *Team FluoroSense*

- Contributed to a **fluorescence-based PFOA detection** concept for **PFAS monitoring**

SKILLS

Data Analysis & Visualization: Python, SQL, Power BI, Advanced Excel

Sustainability & Environmental Analysis: Life Cycle Assessment (LCA), GHG Accounting (Scope 1, 2 & 3), SDG Analysis, Environmental Impact Assessment

Sustainability Tools: openLCA

Laboratory & Characterization: FTIR, UV-Vis Spectroscopy,XRD, SEM, pH Meter, Centrifuge, Autoclave, Membrne Fabrication

Engineering & Simulation: Rhino, Grasshopper, Ladybug Tools

3D Visualization & Design: Revit, SketchUp, Lumion, Enscape, V-Ray, Photoshop, Illustrator

Documentation & Presentation: MS PowerPoint, MS Word, Canva

POSITIONS OF RESPONSIBILITY

Teaching Assistant - LCA Sustainability Analysis and Waste Management | *IIT Hyderabad*

- Prepared structured study materials **LCA** and **sustainability,** supported tutorials and student queries on **LCA methodology**, and evaluated examination scripts against assessment criteria to ensure consistent academic delivery.

Event Design Lead & Coordinator - Svinya 2026 | *Greenko School of Sustainability, IIT Hyderabad*

- **Led and coordinated** the design team, **managing visual identity**, communication materials, and **event execution** to establish the department's inaugural sustainability event as a **recurring initiative.**

Software Club Leader -ArchiTECH Club | *Vastu Kala Academy , GGSIPU*

- **Led and coordinated a 70+ member student** software club, organizing **AutoCAD and Revit** workshops, **managing peer collaboration**, and **driving engagement** through structured training and feedback activities

RELEVANT COURSES

Fundamentals of ESG for Entrepreneurs

- **ESG frameworks**, **Scope 1-3 GHG emissions**, **circular economy**, materiality, biodiversity, and waste management, with exposure to **sustainability reporting, BRSR**, **social** responsibility, and **governance** practices.

Life Cycle Assessment (LCA) for Corporate Sustainability and Climate Change

- Applied **ISO 14040-based LCA methodology**, Life Cycle Inventory **(LCI)**, Life Cycle Impact Assessment **(LCIA)**, and interpretation, with exposure to **GWP, DALY, PDF, and GHG emissions**

Analysis and Methods for Sustainable Engineering

- Studied **multi-objective programming**, **linear/nonlinear optimization**, and sustainability impact factors.
- Implemented **LINGO-based models** to evaluate cost, environmental impacts, water demand, and sustainable system alternatives.

Carbon Capture, Utilization and Storage

- Studied **chemical/physical absorption**, **adsorption**, **membrane separation**, calcium looping, and **direct air capture (DAC)**.
- Analyzed **CO adsorption isotherms**, **BET surface-area data**, relative pressure, and adsorption capacity.
- Applied adsorption data analysis to **adsorbent characterization** and **CCUS** applications.

SWOT Analysis and Risk Management

- Applied **risk assessment**, **FMEA**, portfolio risk analysis, demand uncertainty, inventory risk, and risk pooling.
- Applied **EOQ**, **safety stock**, **correlation analysis**, and **risk mitigation** for **data-driven decision-making**.

INTERESTS AND HOBBIES

Creative & Making: Sketching, painting, origami, Lego building, DIY projects, and organizing & decorating spaces

Exploration & Learning: Travelling, exploring new cultures, meeting new people, learning new software, and hands-on experimentation, Learning new food recipes

Adventure & Fitness: Running, paragliding, rafting, waterfall trekking, and cliff jumping

Curiosity & Design: Exploring stationery brands and material design, science-fiction films (MCU) ,and discovering new ideas